

Negative Literature Answer to the UMST Hilbert-Characterization Question from arXiv:1906.09279

FA/Banach run packet

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Status

This is a literature-already-answered packet. It records that the weaker uniformly micro-semitransitive (UMST) Hilbert-characterization possibility raised in Cabello Sanchez–Dantas–Kadets–Kim–Lee–Martin [1] was answered negatively in the later paper of Cabello Sanchez–Cabello Sanchez [2]. It is not an original proof from this run.

Original Problem

In the final-results section of [1], after proving duality and finite-dimensional restrictions for micro-transitive and UMST spaces, the authors state that no non-Hilbert examples of micro-transitive norms were known and that such norms might all be Hilbertian. They then add that the same could be true for Banach spaces with uniformly micro-semitransitive norm.

The extracted question for this packet is the weaker part:

Does every Banach space with uniformly micro-semitransitive norm have to be Hilbertian?

The source location inspected locally is `data/parsed/arxiv_sources/1906.09279/source.tex`, lines 600–610.

Supporting Answer

The supporting paper [2] studies the semigroup of contractive automorphisms. In Section 4 it defines UMST, cites [1], and explains that Euclidean norms had been the only previously known UMST norms, which made it plausible that UMST might characterize Hilbert spaces. It then says this is false and proves:

Literature Answer. *The UMST Hilbert-characterization question has a negative answer. There are two-dimensional non-Hilbert Banach spaces whose norms are uniformly micro-semitransitive.*

Literature identification. Theorem 4.1 of [2] states that every C^2 norm on the plane whose unit sphere has strictly positive curvature at every point is uniformly micro-semitransitive. Section 4 then gives polar-coordinate examples: if

$$g : \mathbb{R} \rightarrow (0, \infty), \quad 2g'(\theta)^2 + g(\theta)^2 - g(\theta)g''(\theta) > 0$$

is C^2 and π -periodic, then $\|v\|_g = r/g(\theta)$ defines a plane norm satisfying the theorem's hypotheses. The same example section exhibits the non-elliptic curve

$$r = 1 + \frac{1}{17} \sin(4\theta).$$

A two-dimensional Hilbert norm has an ellipsoidal unit ball; this displayed curve is not an ellipse. Hence it supplies a non-Hilbertian Banach plane with UMST norm. $\square$

Scope Limitations

This packet answers only the weaker UMST possibility from [1]. It does not answer the stronger micro-transitive question. In particular, the supporting paper gives contractive automorphisms for the semigroup version, not surjective isometries witnessing micro-transitivity.

Verification Notes

The supporting paper is distinct from the source paper and explicitly cites [1]. Its Section 4 says the proposed UMST characterization is false before proving the positive-curvature plane theorem. The local duplicate check covered arXiv ids 1906.09279 and 2209.04652, plus phrases **micro-transitive**, **uniformly micro-semitransitive**, **UMST**, **Hilbert**, and **contractive automorphisms**. The cheap indexes contained only broader Mazur-rotation entries, not this literature answer.

References

- [1] F. Cabello Sanchez, S. Dantas, V. Kadets, S. K. Kim, H. J. Lee, and M. Martin, *On Banach spaces whose group of isometries acts micro-transitively on the unit sphere*, arXiv:1906.09279.
- [2] F. Cabello Sanchez and J. Cabello Sanchez, *Dynamics of the semigroup of contractive automorphisms of Banach spaces*, arXiv:2209.04652.